COGNITIVE NEUROSCIENCE TOOLKIT

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ICS 496 Capstone Project - Spring 2025

Introduction

This project supports **auditory neuroscience research** in populations unable to engage in complex cognitive tasks. (e.g., infants or individuals with sensory/mobility impairments).

We developed a user-friendly software that allows experimenters to configure **custom auditory experiments** by selecting stimuli and timing parameters.

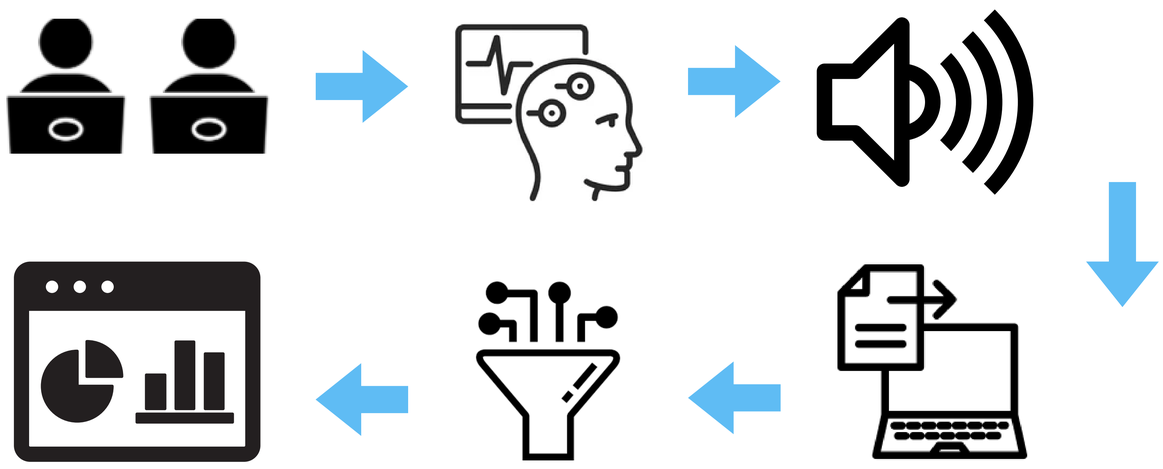
Our software allows researchers to conduct **oddball task** experiments in which a repetitive sequence of "standard" sounds is interrupted by "deviant" sounds to evoke a **Mismatch Negativity** (MMN) response. MMN is a component in **Event-Related Potentials** (ERPs), which reflect the brain's automatic detection of change in sound. MMN offers insight into how quickly and accurately the brain processes auditory changes.

Methodology

- **Preparation:** Assigned team roles to establish clear expectations & responsibilities. Collaboratively dedicated the first 2 weeks to creating a GUI design/mockup and researching MATLAB toolboxes & resources.
  - Salina Thongsisavath: Full-Stack Developer, Project Website Developer, Communications Lead
  - Samantha Mallari: Full-stack Developer, Requirements Lead
  - Adam Bell: Full-stack Developer, Technical Lead
- **Development:** We used GitHub Issues to track and organize ongoing and completed tasks, GitHub repository with automated unit testing for reliability, & GitHub Pages for formally documenting project progress. We also had weekly development meetings to stay on track with our goals.
- **Delivery:** Weekly meetings with our sponsor and her phD student to discuss progress, challenges, and test software.

Solution

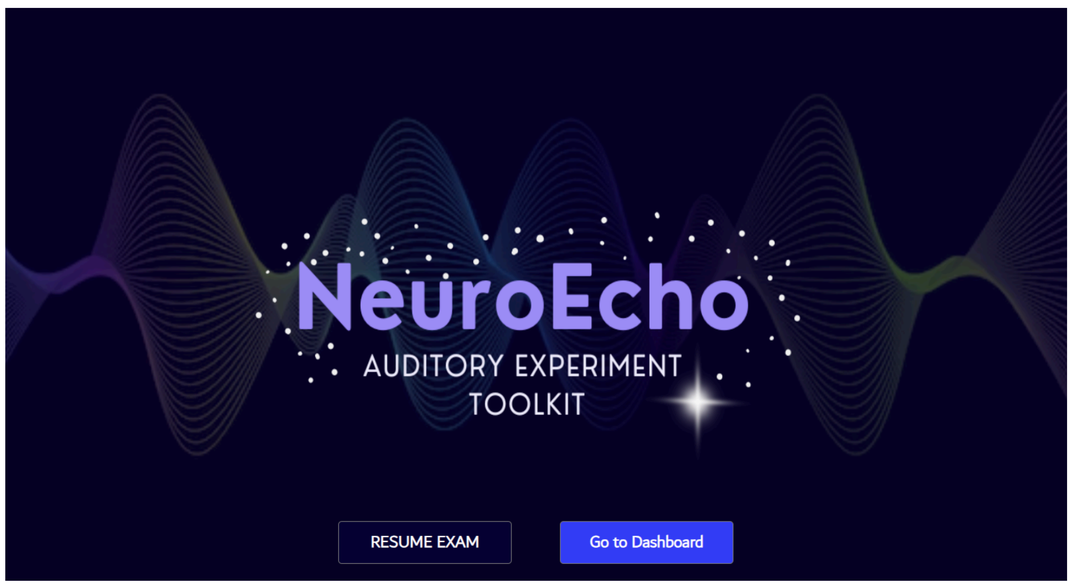
- Real-time auditory/visual stimulus playback.
- Custom GUI to build and save oddball experiments.
- Supports saving in-session annotations as a .txt file.
- Designed to be user-friendly for non-programmers.
- Portable & lightweight — runs on standard laptops for mobile or hospital settings.
- Built in MATLAB, an industry-standard platform for cognitive neuroscience and EEG analysis.



Technology Stack

MATLAB, LSL Lab Streaming Layer, PortAudio Portable Cross-platform Audio I/O, python, PsychToolbox, PsychoPy

Software Preview



Challenges and Learnings

Challenges:

- Needed to deliver the final tool in MATLAB (industry-standard for neuroscience) despite MATLAB App Designer's **limited flexibility**, requiring us to adapt our GUI goals accordingly.
- Ensuring **low-latency** audio playback and precise event triggering under strict timing demands.
- Overcoming PsychToolbox's incompatibility with Apple Silicon by integrating cross-platform alternatives (Lab Streaming Layer, PortAudio)

Learnings:

- Learned to use multiple **industry-standard technologies** for neuroscience research.
- Compiling **dynamic libraries** using CMake.
- **Troubleshooting** third-party open-source libraries.
- Using third-party libraries to send high-precision triggers for data collection.

Acknowledgements

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